

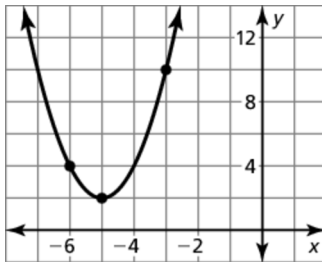
Chapter 2 Test B

2. Let the graph of g be a horizontal shrink by a factor of $\frac{1}{3}$, followed by a translation 1 unit up of the graph of $f(x) = x^2$. Write a rule for g .
3. Let the graph of g be a translation 2 units up and 2 units right, followed by a reflection in the y -axis of the graph of $f(x) = -(x + 3)^2 - 2$. Write a rule for g .
5. Explain why a quadratic function models the data. Then use a linear system to find the model.

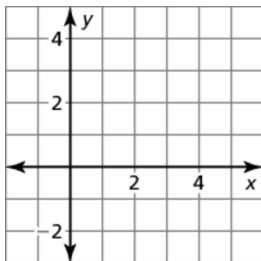
x	2	4	6	8	10
$f(x)$	3	33	87	165	267

Write the equation of the parabola.

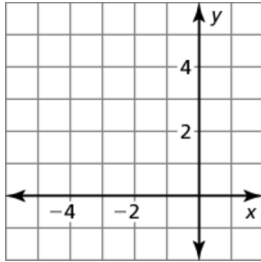
8.



11. Your class council determined that its profit from the upcoming homecoming dance is directly related to the ticket price for the dance. Looking at past dances, the council determined that the profit p can be modeled by the function $p(t) = -12t^2 + 480t + 30$, where t represents the price of each ticket. What should be the price of a ticket to the homecoming dance to maximize the council's profit?
12. Graph $f(x) = -(x - 2)^2 + 4$. Label the vertex and axis of symmetry. Describe where the function is increasing and decreasing.



13. Graph $g(x) = \frac{1}{2}x^2 + 2x + 4$. Label the vertex and axis of symmetry. Describe where the function is increasing and decreasing.



15. A biologist took a count of spotted trout that migrate to the south end of a lake during the winter months. The table shows the population count at the south end of the lake after week 10.

Week, x	2	4	6	8	10
Population, y	185	209	217	209	185

Use quadratic regression to estimate the number of spotted trout at the south end of the lake after week 14.